Practical Ai Collaboration

- Prompt engineering strategies

The Jackson Laboratory

- Ai **Fluency**: the 4D Framework
- Why prompt engineering matters
- Structure of effective prompts
- LLMs in research

Prompt Engineering

The 4D Framework

The 4Ds (Anthropic)

Delegation → Description → Discernment → Diligence

The 4Ds (Anthropic)

Prompt Engineering lives here

Delegation → Description → Discernment → Diligence

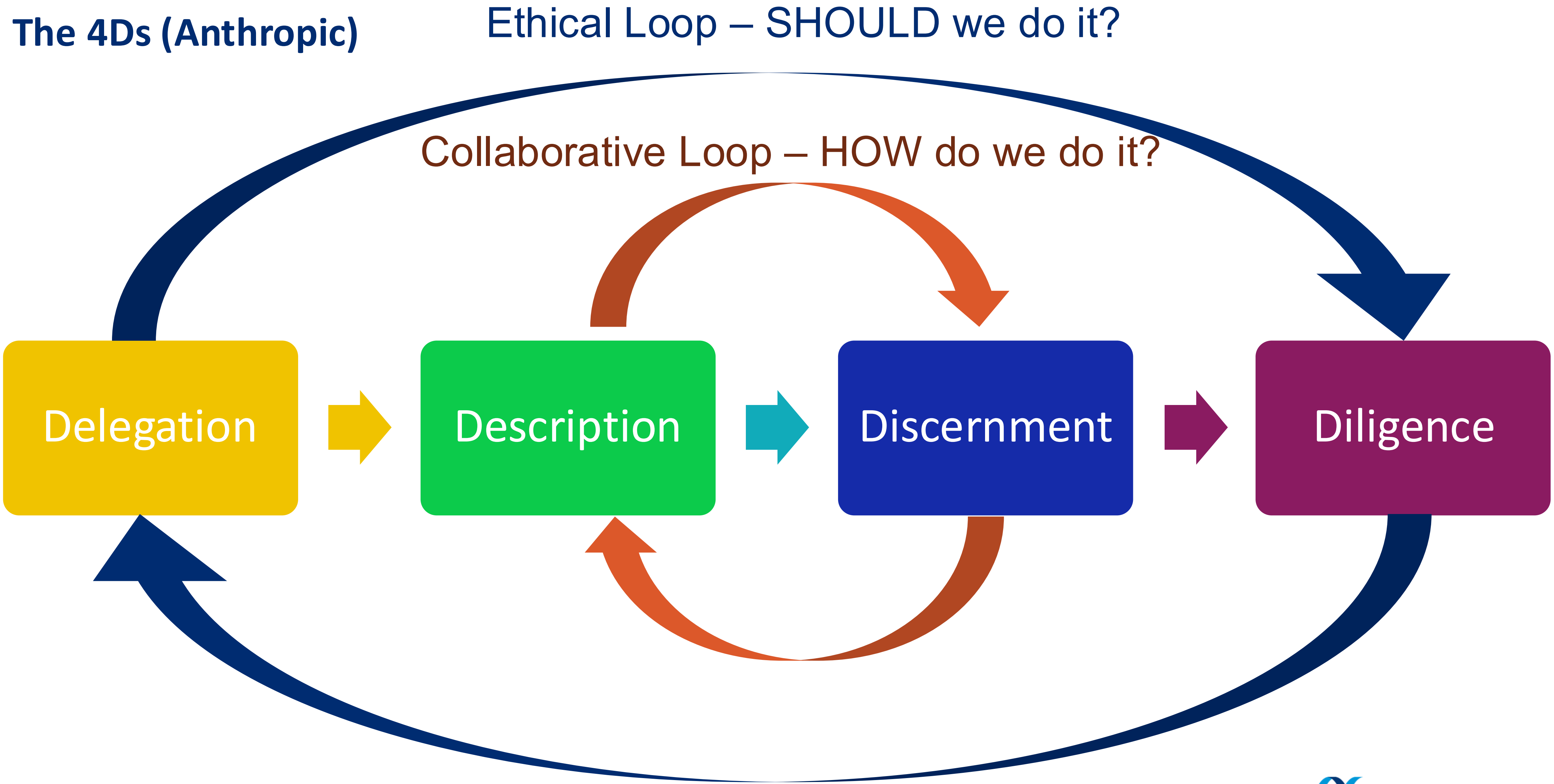
- Problem Awareness
- Platform Awareness
- **Task Delegation**

- **Context**
- Product
- Process
- Performance

- Built better rubrics
- **Make expert thinking visible**
- Name behavioral norms
- Map disciplines key products
- Diagnose failures

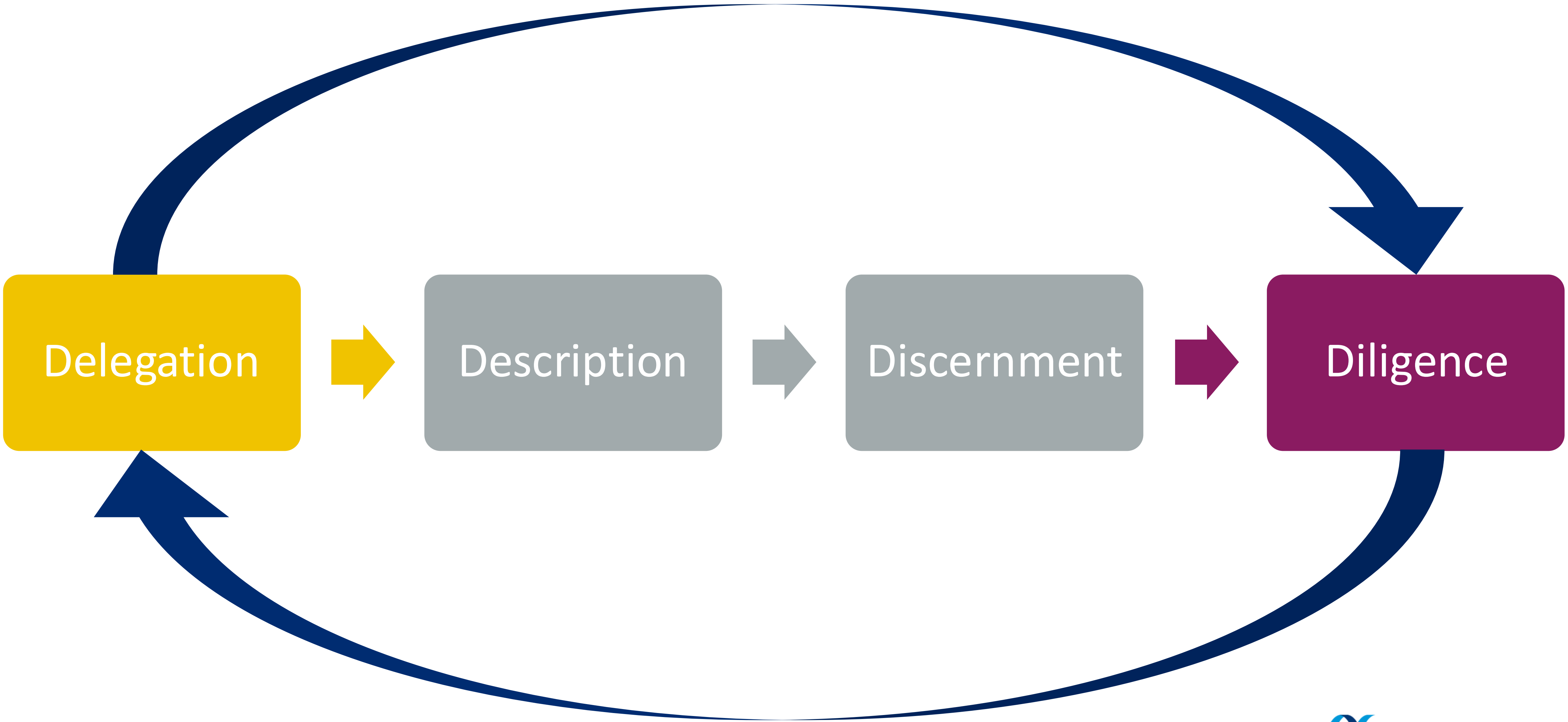
- Codify **ethics**
- Clarify
- Transparency

The 4Ds (Anthropic)



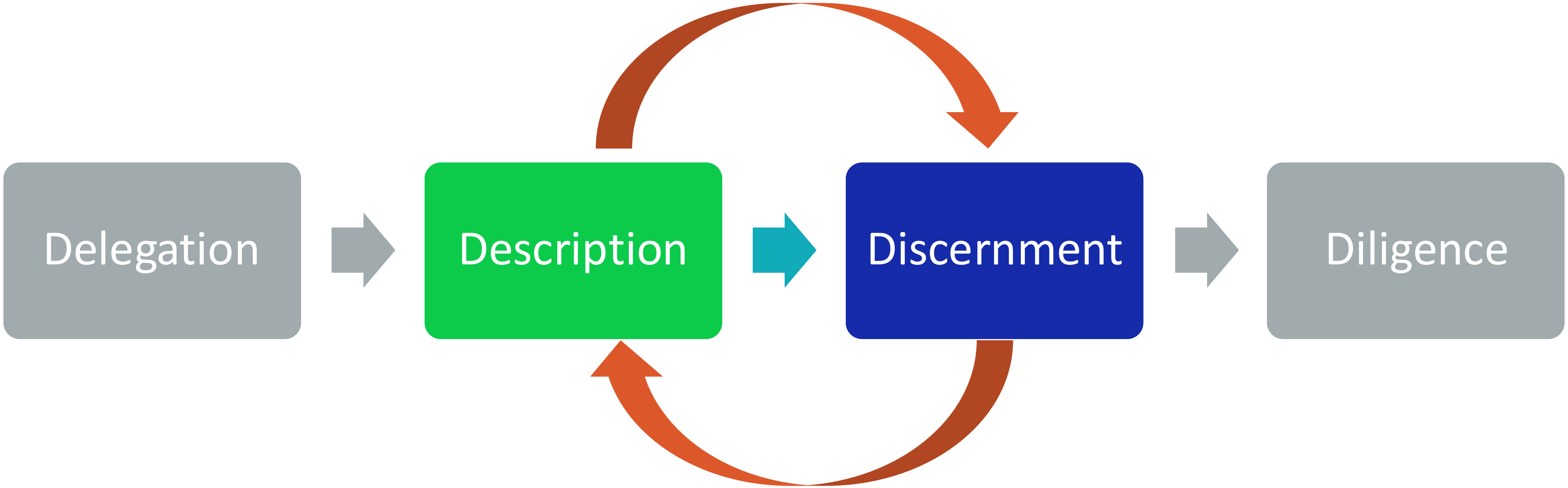
The 4Ds (Anthropic)

Ethical Loop – SHOULD we do it?



The 4Ds (Anthropic)

Collaborative Loop – HOW do we do it?



Delegation Mapping

Should I use Ai here? Discussion

Delegation mapping

		Human Prompter Domain Expertise	
		Low expertise	High expertise
Outcome	High stakes	CAUTION ZONE • Human oversight essential	• **Human Judgment first** • Ai in support role
	Low stakes	Strong delegation candidate • Ai completes, • Human verifies	Strong delegation candidate • Ai drafts, • Human discernment

When should scientific trainees use Ai?

User expertise level

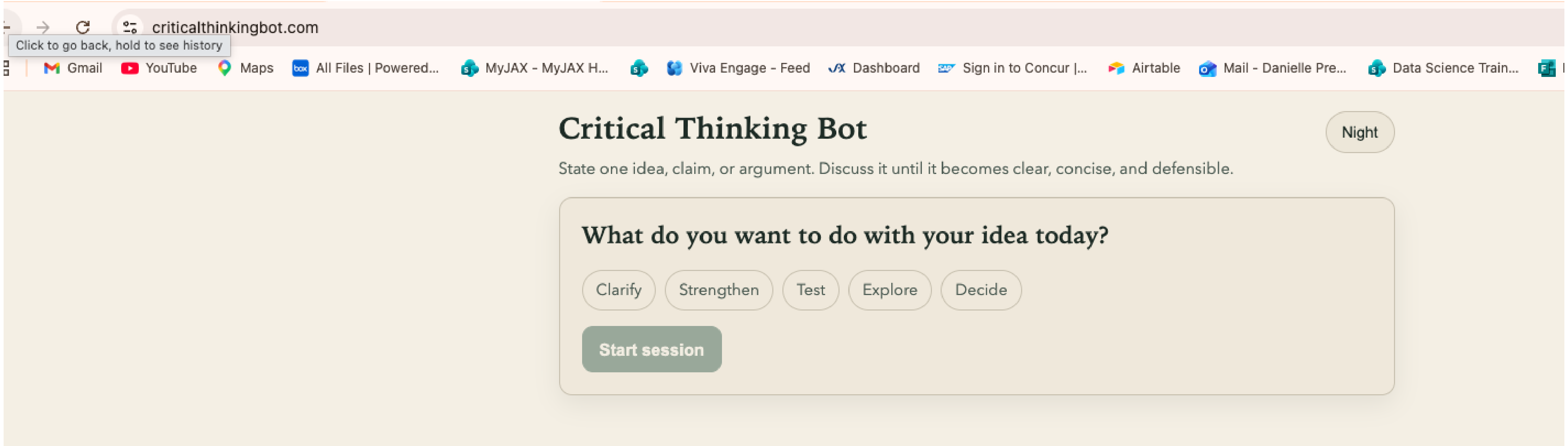
Type of automation		Trainee (developing expertise)	Expert (established expertise)
	Mechanical (executing established procedures)	***Examples:*** Statistical software running chosen tests, search engines finding the right information ***Effect:*** Enables focus on cognitive skills ***Training impact:*** Positive/Neutral (preserves learning)	***Examples:*** Automated computational pipelines, batch processing, data management tools ***Effect:*** Increases productivity
	Cognitive (reasoning, synthesis, judgment)	***Examples:*** GenAI writing analysis code, GenAI interpreting results, GenAI synthesizing literature ***Effect:*** Bypasses skill development ***Training impact:*** NEGATIVE (prevents expertise development)	***Examples:*** GenAI for exploring variations, AI-assisted iteration, augmented workflows ***Effect:*** May augment capability

Figure 1 from Krishnan 2026a

Why Prompt Engineering Matters

The Critical Thinking Bot

- Socratic method



- You choose a mode: clarify, strengthen, test, explore, decide
- Asks questions instead of offering conclusions

The Critical Thinking Bot

Dimension	ChatGPT or Claude (generic)	ChatGPT or Claude (good construction)	Critical Thinking Bot
Output type	Informational Summary or essay	Precisely scoped analysis matching your spec	Socratic questions and reflection prompts
Directionality	Flows toward you (gives you content)	Flows toward you	Flow back (asks you to examine your claim)
What it produces	Fluent text: facts, arguments, pro/cons	Targeted text: the format and depth you defined	Questions: gaps, assumptions, overlooked angles
Cognitive role for user	Reader/Evaluator	Director/Collaborator	Examiner/defender
Best used for	Research, drafting, summarizing, exploring	Analysis, writing, ideation with constraints	Testing arguments, clarifying belief

What is prompt engineering?

Prompt Engineering IS…	Prompt Engineering IS NOT….
• Structuring communication with precision • Defining scope, role, output format • Iterating based on what the model understood and **misunderstood** • A learnable, transferable skill	• Finding a magic phrase or formula • Manipulating the model • One-time set up that you never re-visit • Guaranteed to produce perfect results

Good prompts for augmentation don't get you answers, they get you better questions.

Why it matters

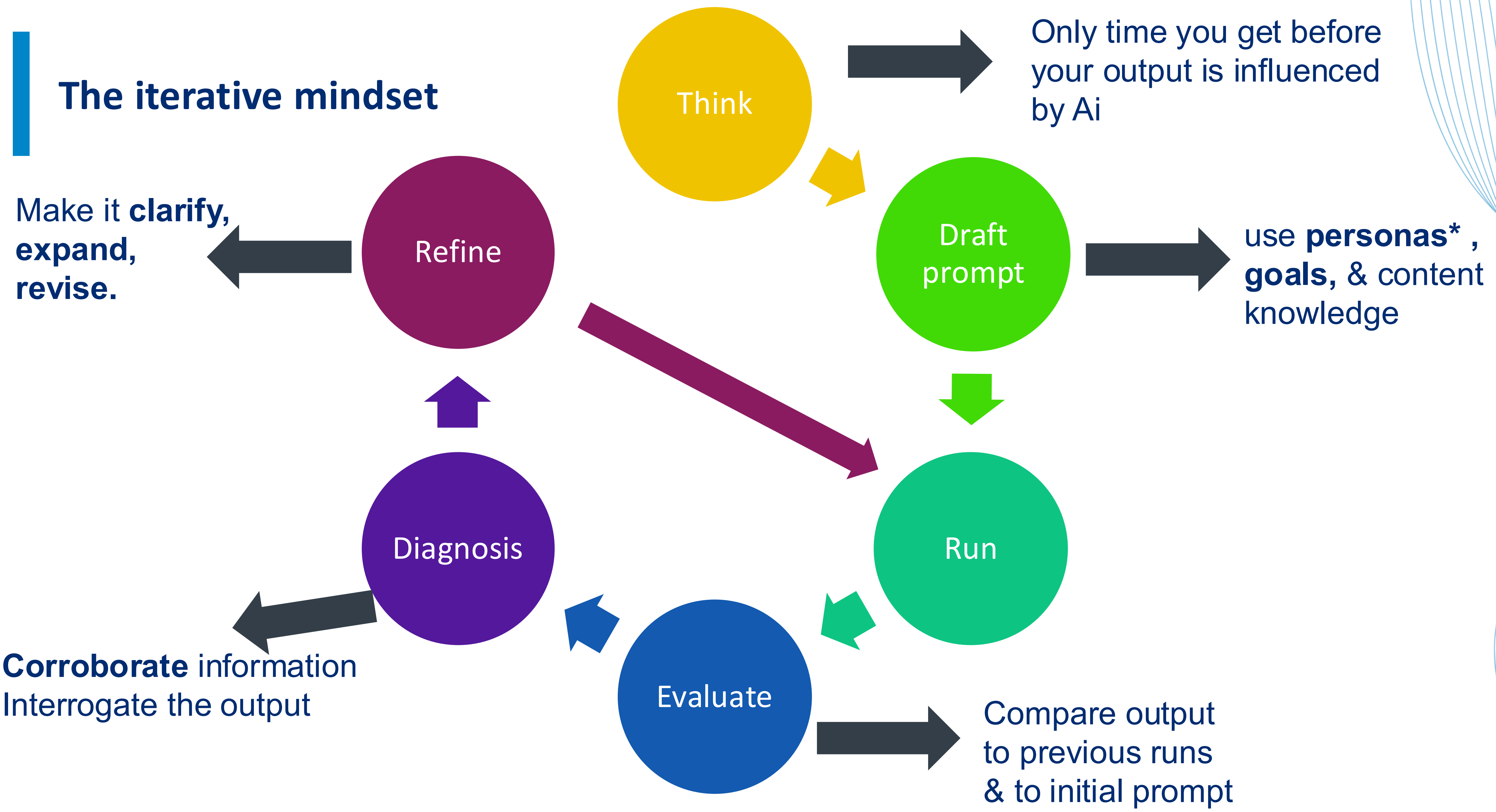
	Vague Prompt	Precise Prompt
Input	“Tell me about leadership”	“List 5 evidence-based leadership behaviours that improve team psychological safety, with one sentence explaining each. Write for a mid-level manager audience”
Distribution	Wide: could produce a history essay, a listicle, a philosophical reflection, a definition, or a how-to guide	Narrow: role defined, format defined, audience defined, scope defined. **We will use the 3 Ps as a guide….**
Output	Fluent, but generic; may sound good but misses your actual need	Targeted, usable, appropriately scoped

The iterative mindset

1. There is no perfect prompt that will save you: Prompting is diagnostic
    - Effective prompting is not about find the right magic words
2. When a prompt doesn't work, your job is to ask
    - What did the model **think** I was asking for? What **assumptions** did the model make?
    - What was **ambiguous**? What constraint was **missing**? What **context** did I fail to provide?
3. Each iteration teaches you about the model and the clarity of your own thinking

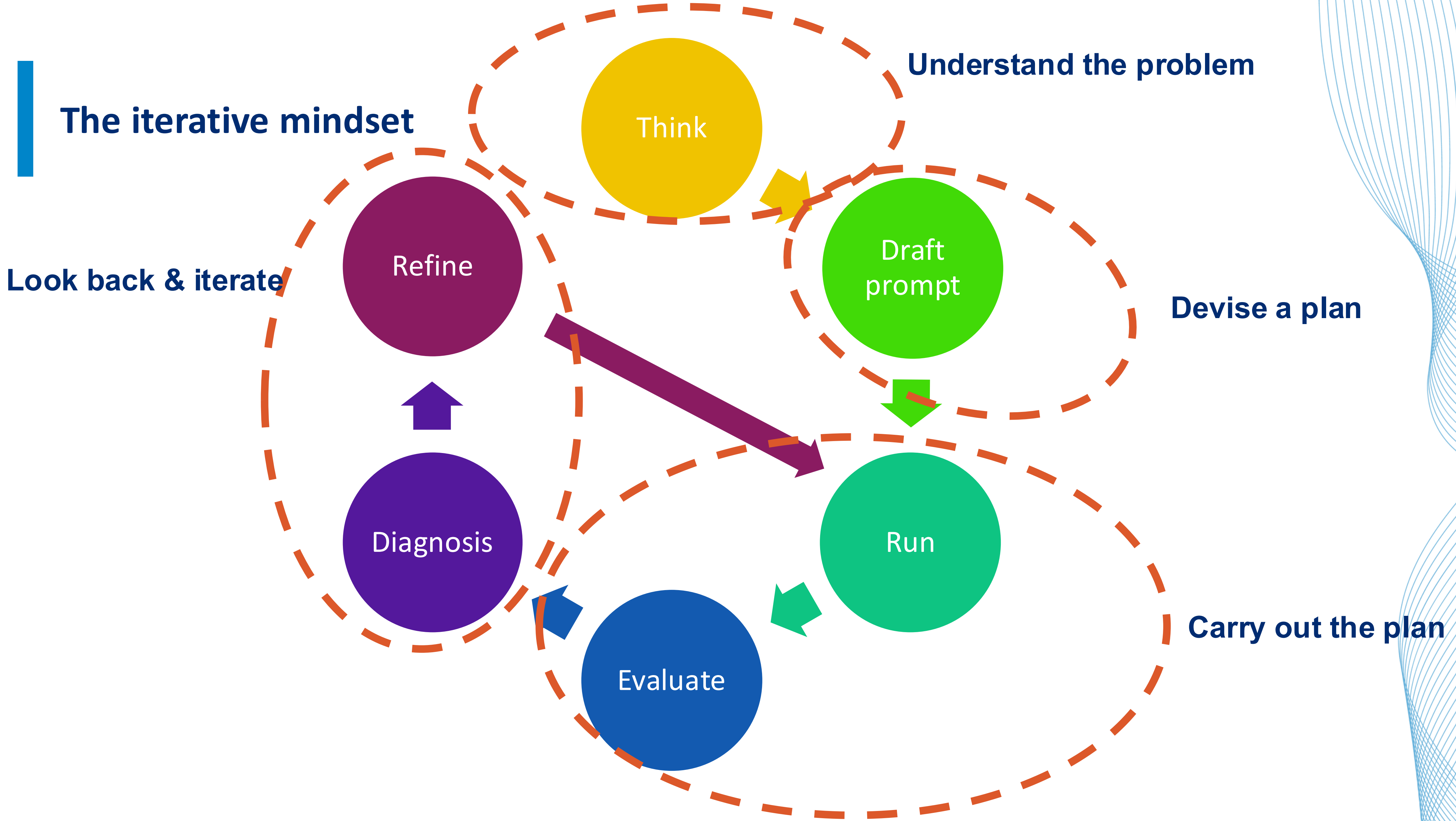
Draft prompt → **Run** → **Evaluate** the output → **Diagnose** the gap → **Refine** → **Re-run & Compare**

The iterative mindset



George Polya's Four-step Problem solving process (1945)	Prompt Manifestation
Understand the problem	• Clearly define your goal • Identify known/unknowns • Restate the problem • If possible: **draw out the problem** • Provide necessary **context**
Devise a Plan	• Prompt **strategy:** few-shot, chain-of-thought • Break down complex task • Define constraints & Expectation
Carry out the Plan	• Submit prompt • Refine prompt • **Document** prompt & output
Look back & Reflect: Iterate	• **Evaluate** & verify output • Identify areas for improvement

The iterative mindset



The iterative mindset

Draft prompt → **Run** → **Evaluate** the output → **Diagnose** the gap → **Refine** → **Re-run & Compare**

Common failures

Symptom	Likely Cause	Fix
Too long/too much	No length or format constraint	Add in “3 bullet points” or “in 150 words”
To generic/shallow	No audience or depth specified	Specify audience, expertise level or use case
Wrong tone	No tone or role instruction	Add “Write for a skeptical executive” or ”use plain language”
Hallucinations	Model filling gaps with plausible-sounding content	Add sources (RAG), limit scope, ask model to flag uncertainty
Missed the point	Ambigous core request	Re-write the core verb – be explicit about the action you want

[Submitted on 5 Dec 2025]

Prompting Science Report 4: Playing Pretend: Expert Personas Don't Improve Factual Accuracy

Savir Basil, Ina Shapiro, Dan Shapiro, Ethan Mollick, Lilach Mollick, Lennart Meincke

This is the fourth in a series of short reports that help business, education, and policy leaders understand the technical details of working with AI through rigorous testing. Here, we ask whether assigning personas to models improves performance on difficult objective multiple-choice questions. We study both domain-specific expert personas and low-knowledge personas, evaluating six models on GPQA Diamond (Rein et al. 2024) and MMLU-Pro (Wang et al. 2024), graduate-level questions spanning science, engineering, and law.

We tested three approaches:

-In-Domain Experts: Assigning the model an expert persona ("you are a physics expert") matched to the problem type (physics problems) had no significant impact on performance (with the exception of the Gemini 2.0 Flash model).

-Off-Domain Experts (Domain-Mismatched): Assigning the model an expert persona ("you are a physics expert") not matched to the problem type (law problems) resulted in marginal differences.

-Low-Knowledge Personas: We assigned the model negative capability personas (layperson, young child, toddler), which were generally harmful to benchmark accuracy.

Across both benchmarks, persona prompts generally did not improve accuracy relative to a no-persona baseline. Expert personas showed no consistent benefit across models, with few exceptions. Domain-mismatched expert personas sometimes degraded performance. Low-knowledge personas often reduced accuracy. These results are about the accuracy of answers only; personas may serve other purposes (such as altering the tone of outputs), beyond improving factual performance.

Key Take-aways

- Prompt engineering is **not** a one-time trick
- Prompting is your primary interface for using Ai as a ‘thought partner”
- Your prompt is context:

  vague input = wide probability distribution for next token = fluent but unfocused
- Effective prompting is **iterative**. Diagnose why it failed.
- Different tools (Socratic bot versus LLM) have different output goals and will be useful at different points in your workflow.

Appendix

Element	What to specify	Example
Role	Who should the AI be?	"Act as a skeptical editor reviewing this for a business audience."
Task	What specific action?	"Summarize" not "Tell me about"
Context	Background the model needs	"This is for a board presentation to non-technical stakeholders."
Constraints	Scope, length, what to avoid	"Under 200 words. No jargon. Do not include implementation details."
Format	How should output be structured?	"Return as a numbered list with one sentence per item."
Audience	Who will read this?	"Write for a manager with no data science background."
Examples	Show don't tell	"Here's a strong example of the tone I want: [paste example]"

The Three Ps

The Details of an effective prompt (Description)

The Three Ps: Product, Process, Performance

Product:

what output do you want?

Process:

How should Ai approach the task?

Performance:

How should the Ai behave?

The Three Ps: Product, Process, Performance

Product:

- Be specific
  - length, tone, **format, audience**
- Example:
  - Summarize this paper

    vs
  - *Summarize this paper in three bullet points for a clinician with no statistical background, focusing solely on clinical implications*

Process:

- Break complex tasks into steps
- Specify the reasoning approach
- Example:

  Before answering, identify the key assumptions in the question. Then, give your answer step-by-step.

Performance:

- Define role, confidence level, and what to do when uncertain
- Example:

  You are a skeptical peer reviewer. If you are uncertain about a claim, say so explicitly rather than guessing.

The Three Ps: Product, Process, Performance

Six practical tips:

1. Give **context** – Ai can’t read your mind…yet
2. Break complex tasks into steps: chain-of-thought prompting
3. Show examples: zero-shot (no examples), one-shot (1 example), few-shot
4. Specify constraints: length, format, what to exclude, what not to do!
5. Ask Ai to think first since this forces process over pattern-matching
6. Define tone or role

Bad Prompt Make-over

Activity

Pick one of the following prompts and work together

1. "Tell me about CRISPR."
2. "Write me a methods section."
3. "Is this result significant?"
4. "Summarize the literature on depression."
5. "Help me with my grant."

Example of a bad prompt make-over

Summarize the major mouse genetics discoveries published in 2021 regarding the genes *Mxrl2*, *Tnfip7*, and *Cbln4r3*. Include each gene’s chromosomal location, knockout phenotype, and any behavioral abnormalities reported. Present the findings confidently and in detail.

Example of a bad prompt make-over

I want to ask about mouse genetics studies, but the gene symbols I'm giving you may be fictional or may not exist in any database. Before providing any summary, do the following:

Check whether *Mxrl2*, *Tnfip7*, and *Cbln4r3* correspond to real mouse genes.

If they are not real, clearly state that no validated data exists.

If information is missing or uncertain, explain the gap rather than filling it in.

Provide safe, general guidance on how such mouse genetics studies *would* typically be structured, without inventing data.

Another example of a bad prompt make-over

“Summarize TP53”

Another example of a bad prompt make-over

- **Poor Prompt:**

“Summarize TP53”

- **Better prompt:**

“Summarize the role of TP53 in DNA damage response in ≤ 3 bullet points, citing key papers”

- **Even Better prompt:?**

Even excellent prompting cannot fix G.I.G.O. principle

- Bad data
- Unclear goals
- Weak domain knowledge
- Poor experimental design
- Misaligned incentives

Ai in Research

Context is key

Bioinformatics Use Cases

1. “vibe coding”

- “There's a new kind of coding I call ‘vibe coding’, where you fully give in to the vibes, embrace exponentials, and forget that the code even exists.” ~ **Karpathy, 2025**

2. Research Pipeline

Delegation mapping

		Human Prompter Domain Expertise	
		Low expertise	High expertise
Outcome	High stakes		
	Low stakes		

Literature Review → Experimental Design → Data Collection → Data Analysis → Writing

Bioinformatics Use Cases

1. **“vibe coding”**
2. **Research Pipeline**
   - ~~**Prompt Engineering:**~~ **Context Engineering: Augmentation, analysis,**
     - Summarizing literature, identifying, and predicting genetic variants, hypothesis generation
   - **Automation**
     - creating workflows, annotating genetic variants

- Remember:
  - Human-in-the-loop and reproducibility
  - **“AI can’t replace thinking, but it can amplify poor thinking.”**

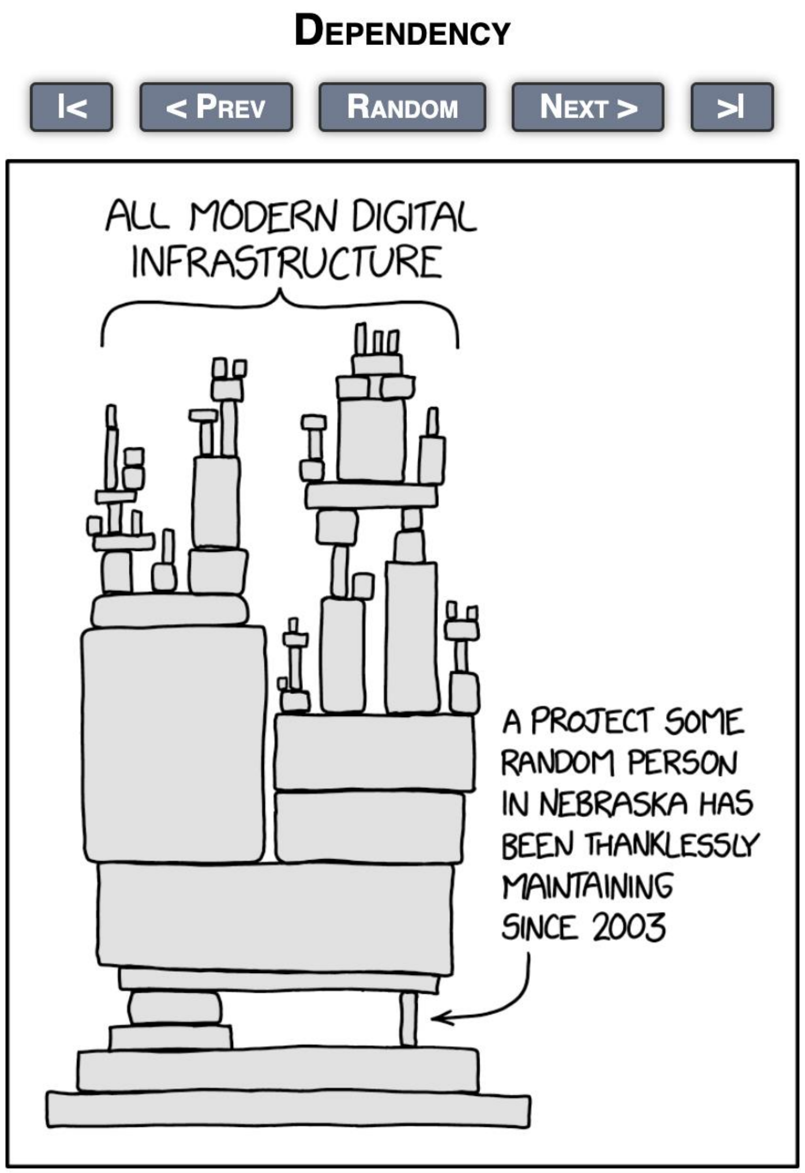
https://www.promptingguide.ai/guides/context-engineering-guide

"vibe coding"

- A coding style: you converse with an LLM, accept most suggestions, and iterate quickly without worrying about syntax
- **Use case:** rapid prototyping
- **Example:** Gemini in Colab

The human still needs to:

1. **Troubleshoot**
   - Have LLM create use cases and try to break your code
2. **Refactor**
   - Reduce tech debt
   - streamline
   - Ensure repeatability and reproducibility



https://xkcd.com/2347/

A team of Rivals: the LLM Council

https://github.com/karpathy/llm-council

A team of Rivals: the LLM Council

1. **The Contrarian**
2. **The First Principles**
3. **The Expansionist**
4. **The Outsider**
5. **The Executor**

Prompt Engineering for Research

How to create a robust prompt

- **Context**
  - Set up your account (settings)
  - Writing an effective “preamble”
  - Principles of writing an aligned task
- **Reproducibility** : Ensure that you are saving your prompts
- **Strategy:** zero shot versus few shot
- **Risks**: Alignment

Prompt Engineering for Research

Clear, contextual instructions + examples = accurate, reproducible output

Besides defining the actual task, "content", the structure and style of your prompt plays a significant role in guiding AI's response.

Pattern	Use Cases	Example	Risk
Zero-Shot	General	Translation/summary "Is this variant pathogenic or benign?"	Ambiguity
Few-shot	Style Mimic	2-3 demos Add examples with ClinVar annotation, and a few covering unusual cases (missense, splice variations etc.)	Token cost
Chain-of-thought	Reasoning	"explain step by step"	Long/slow

Prompt Engineering for Research

Clear, contextual instructions + examples = accurate, reproducible output

Preamble or Persona:

- Uncertainty instructions
- Request attribution

Prompt format should prioritize:

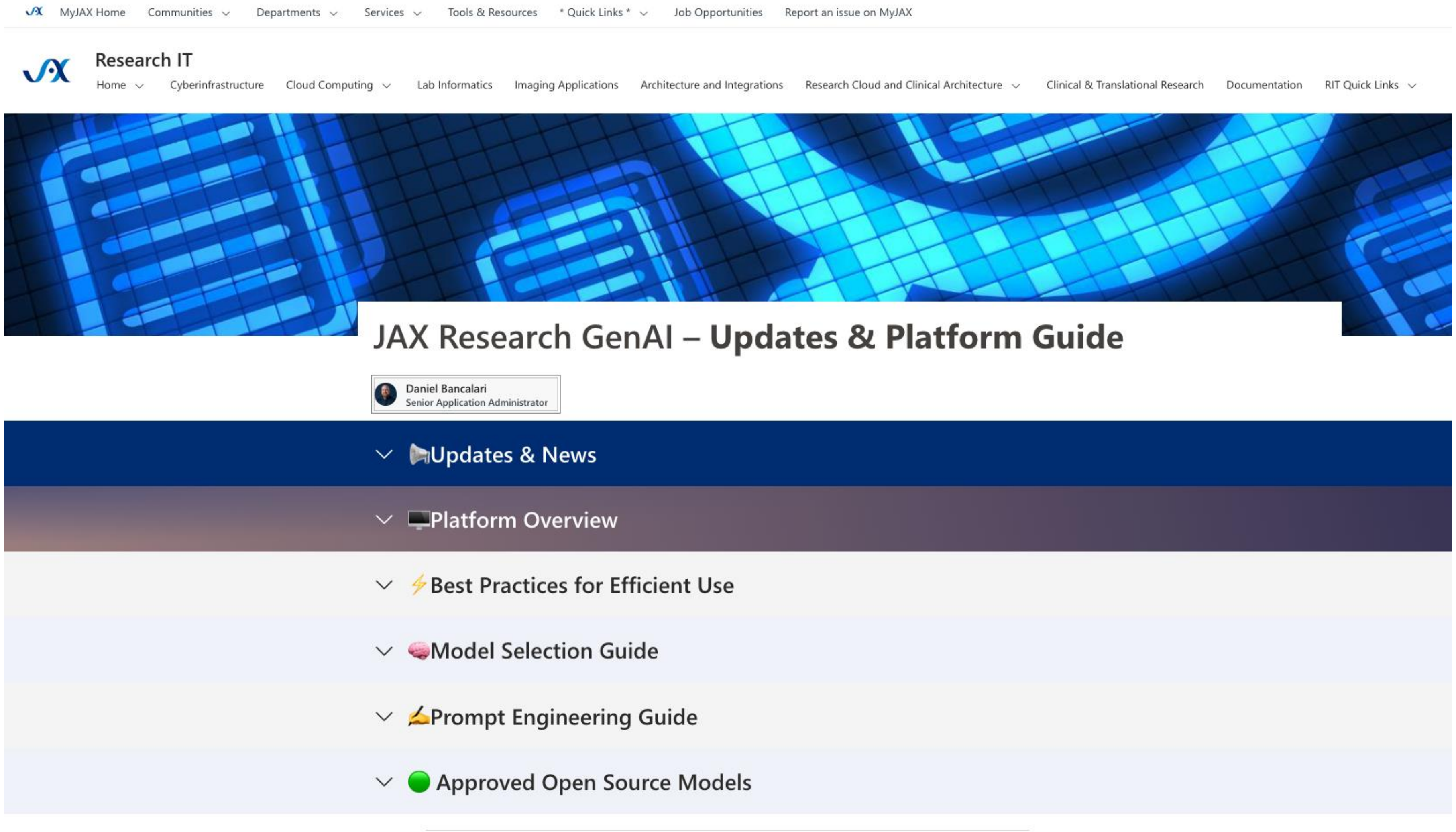
- Simplicity of task
- Specificity
- Examples
- Iteration

https://cloud.google.com/vertex-ai/generative-ai/docs/learn/prompts/introduction-prompt-design#task

Simplicity	• Clearly set task, with content parameters	" Your objective is to help students with math problems without directly giving them the answer" 1. Understand the problem 2. Understand where the student is stuck 3. Give a hint for the next step of the problem
Specificity	• Note any output constraints • Jupyter notebooks? Python? R? • Set Persona/tone • Elevator pitch or technical presentation	You are a math tutor, here to help students with their math homework. Do not give them the answers, only hints. Respond in a casual and technical tone.
Examples	• Context, and background information • Few-shot examples; leverage CoT to see how they got their answer • Integrate external tools	1. Provide a lesson plan 2. Give examples of what the response should look like
Iteration	• Adjust temperature • closer to 0 → rigid protocol • closer to 1 → exploratory	

https://cloud.google.com/vertex-ai/generative-ai/docs/learn/prompts/prompt-design-strategies#sample-prompt-template

JAX Research IT has a website dedicated to it!



https://jacksonlaboratory.sharepoint.com/sites/ResearchIT/SitePages/JAX-Research-IT-%E2%80%94-Generative-AI-Updates.aspx

Discernment

Evaluation

Rapid Evaluation & Prompt-Debug Checklist

Quick Metrics

- Accuracy (yes/no tasks)
- Precision & Recall (classes)
- BLEU / ROUGE (text quality)
- Human spot-check & citations

Prompt-Debug Steps

1. Clarify task & style
2. Provide context + examples
3. Adjust temperature / top-p
4. Iterate & re-evaluate

Discernment (& Evaluation)

- Domain expertise necessary (Human-in-the-loop) → can a human verify it
- General Checklist:
  - **Factual accuracy**
    - Citation validity, facts being asserted
  - **Completeness & relevance**
    - Omitted alternatives
  - **Verifiable**
  - **Uncertainty handling**
    - Hidden assumptions
  - **Reproducibility**
    - Does the same prompt give similar answers?

Diligence

Before moving onto Ethics (tomorrow)

1. **Fairness**
   - Does this output treat all groups equitably?
   - Does training introduce bias?
2. **Accuracy**
   - Have you verified the claims independently?
   - Is the output citable?
3. **Transparency**
   - Have you disclosed Ai involvement
4. **Accountability**
   - When Ai output is wrong, who is responsible?